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Culture-dependent isolation and morpho-biochemical characterization of bacterial populations associated with the gut of litchi stink bug, *Tessaratoma javanica* Hemiptera (Thunberg)

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Abstract- The insect gut is the largest reservoir of diverse microbial communities. The bacterial-insect interactions range from parasitism to mutualism. These microbes can be harmful or beneficial to the host. Symbiotic bacteria and their insect hosts depend on each other in various ways and play crucial roles in nutrition, digestion, immunity, detoxification, development, and evolution. The Litchi stink bug, *Tessaratoma javanica*, is a significant pest affecting *Litchi chinensis* Sonn., as well as other plants such as Longan, Kusum, and Mulberry. This study aimed to evaluate the cultivable gut-associated bacteria across different developmental stages of *T. javanica*. Bacteria were characterized through morphological, biochemical, and molecular methods. A total of 24 different bacterial colonies on the basis of colony morphology were isolated on three different cultural media: Peptone Yeast Extract Agar (PYEA), Nutrient Agar (NA), and Brain Heart Infusion Agar (BHI). The highest number of isolates was obtained on nutrient agar Out of total, 10 different bacterial species belonging to 7 different families. Most isolates were Gram-positive and rod-shaped. The different bacteria species viz., *Microbacterium paraoxydans, Bacillus tequilensis, Bacillus* sp., *Staphylococcus arlettae, Staphylococcus warneri, Microbacterium testaceum, Enterobacter, Pseudomonas* sp., *Bacillus subtilis,* and *Acinetobacter* were identified to be associated with different developmental stage. The predominant bacteria belonged to the family Bacillaceae. Further, genera such as *Microbacterium paraoxydans* bacterial species belongs to the family Microbacteriaceae was present across all developmental stages. Adult stages exhibited higher bacterial counts compared to nymphal stages. This study provides the first detailed description of bacterial isolates associated with different growth stages of *T. javanica*. The findings may facilitate the development of targeted bacterial management strategies against the litchi stink bug in both laboratory and field settings.

Keywords: *Litchi chinensis*, Symbiotic bacteria, Host fitness, Developmental stage, *Tessaratoma javanica*

INTRODUCTION

Throughout their evolutionary history, insects have been known to be associated with microbes such as bacteria and fungi. Certain bacterial species are referred to as endosymbionts as they reside in specialized cells within the insects.[1] The interaction between insects and bacteria is a complex and multidimensional relationship, and the bacterial species present within an insect gut may exhibit mutualistic, commensal, or even pathogenic behaviors.[2] Insects' physiological as well as biochemical adaptations have aided in their ability to adapt to different

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environmental niches.[3] Insect gut microbiota is known to provide immunity and protection against a variety of predators, pathogens and parasites, compensate for a diet lacking in nutrient specially in case of sap sucking insects, mediate inter- and intra-specific communication, regulate mating and reproductive success, aids digestion and detoxification, provide essential amino acid and metabolic compounds.[4-9] Although symbionts also affect hot fitness and pesticide sensitivity[10,11] and providing vital nutrients, which is followed by digestion and detoxification. This suggests that there is an interactive relationship between gut microbiota and insect host[12], which play a crucial role in development, morphogenesis, immunology, behavior, and metabolism, and regulate life cycle functions[13].

Numerous biotic and abiotic factors affect the development and maintenance of these microbial communities within the insect host, influencing their diversity and composition.[8] Diet, environment, and host species are some of the main variables influencing the makeup of symbiont bacterial communities in insects. In insects, host species, environment, and diet are among the main variables influencing the composition of the symbiont bacterial population. Specialized insects, such as herbivores and blood-feeders, have unique microbiota that are adapted to meet their dietary requirement.[14] Microbial diversity is also greatly influenced by habitat factors such as temperature, humidity and exposures to environmental microorganisms.[15] Furthermore, it has been demonstrated that endosymbionts influence their hosts' heat tolerance[16], make it easier to employ new hosts[17]. Researchers reported that certain holometabolous insects go through four stages of development, including egg, larva, pupa, and adult.

Tessaratoma javanica (Thunberg), also known as Litchi stink bug, belongs to the order Hemiptera: Tessaratomidae (family) is an economically sucking pest of *Litchi chinensis* Sonn. (Sapindacea).[18] It is a major pest of litchi but beside litchi it has a wide range of host plant such as Longan, Pomegranate, Pummelo, Mahua, Kusum, Rambutan, Castor and Eucalyptus, and Loquat[18,19] and has also been reported a pest of apple, pear, mulberry plant *Morus alba*. This stink bug is found in different parts of the world, including China, Australia, Thailand, Myanmar, Vietnam, the Philippines, and India. In India, it was reported in the Chotanagpur plateau of Jharkhand, Bihar, and Chhattisgarh.[18-21]

The bacterial communities associated with *Tessaratoma javanica* have not been thoroughly studied using either the culture-dependent or culture-independent methodologies. The characterization of the cultivable gut bacterial community across the several developmental stages of *T. javanica* has not been previously published, which is noteworthy. Using morphological and biochemical techniques, the current study aims to discover and describe these gut-associated bacterial populations. For future studies on the function and role of bacterial communities in *Tessaratoma javanica* development and management, the result will serve as a baseline.

MATERIALS & METHODS

Collection of samples

Collection of specimens from orchids of litchi trees at the Research Complex for Eastern Region Farming System Research Centre for Hill and Plateau Region (ICAR, RCER, FSRCHPR), Namkum, Ranchi, 23º 45´ N latitude and 85º 30´ longitude, having an elevation of 620m above mean sea level (AMLS), India, during the year 2022. Different specimens represent the different stages of *T. javanica,* which include the 2nd instar nymph, 4th instar nymph, and an adult Male and Female. Specimens of different stages, nymph and adult, were collected from the field and brought in a transparent plastic bag alive to the laboratory and stored in a refrigerator with having temperature of -20ºC for 5-10 min before dissection. The identifications of specimens were conducted using morphological features as described by Praveen *et al.* (2015)[22].

Dissection and isolation of the gut

Different stages, including 2nd nymph, 4th nymph, and adult male and female of *T. javanica* stink bug, were selected for the isolation of gut bacterial isolates following the protocol described by Naaz *et al.* (2016)[23] with minor modifications. Before dissection, insect samples (2nd instar nymph, 4th instar nymph, male adult and female adult) were kept at -20°C to anesthetize it for 5-10 minutes and then Insect samples were surface sterilized with 70% ethanol for 30 s followed by 0.1% mercuric chloride and then washed 3 times with sterilized distilled water to remove any external surface contamination. The 4th instar nymph and adult male and female stink bug were dissected aseptically in a Petri plate filled with PBS with the help of sterilized scissors, forceps, and tweezers under a laminar

air flow hood. Firstly, the wings and back plates were removed from the stink bug, and then dissected its ventral side was dissected and then remove the midgut. The 2nd instar nymphs were used as whole bodies for processing. After dissection, the gut of different stages was placed in an Eppendorf tube containing 500 µl PBS (Phosphate buffer solution) and homogenized with the help of a sterilized micro pestle. Serial dilution was done up to 10-4 in 1000µl. To isolate bacteria, gut contents were first spread with the help of a sterilized L-shaped glass spreader on a sterilized petri plate having three different media, Nutrient agar (NA), Peptone yeast extract agar (PYEA) and Brain heart infusion agar (BHI). The Petri plate was then kept in an incubator for 24-48 hours at 37±2°C for bacterial colony growth. After 24-48 hours, mixed types of bacterial growth were observed in different media on the plate. The different bacterial growth was then streaked into a zig-zag pattern with the help of a sterilized inoculation loop into a Petri plate having 3 different types of agar media and kept for 48-72 hours in an incubator for bacterial growth. The pure colony of bacterial growth was observed by repeated sub-culturing. The pure colony was then preserved in NA slant at 4°C, and pure bacterial growth was maintained in 50 % glycerol stocks at -20°C.

Identification and Characterization of Bacteria

a. Morphological (color, shape, and size, and Gram staining)
b. Cultural (growth of bacteria on broth medium)
c. Biochemical test.

Biochemical tests include: Citrate, Catalase, Oxidase, Indole tests, and Triple Sugar Iron (TSI) test.

Carbohydrate fermentation test includes: D- D-Glucose, D- D-Glucose (gas production), Sucrose, Lactose, Maltose, Mannose, and D- Xylose

The characterization of pure culture was done by standard techniques, and the results were compared with Bergey's Manual of Determinative Bacteriology.

RESULTS

Table 1: Isolation of gut bacteria from different developmental stages of *Tessaratoma javanica*

Developmental stages of *Tessaratoma javanica* used for bacterial isolation	Culture media used					
	Peptone Yeast Extract Agar (PYEA)		Nutrient Agar (NA)		Brain Heart Infusion (BHI)	
	No. of isolated bacterial colonies	Bacterial isolate number	No. of isolated bacterial colonies	Bacterial isolate number	No. of isolated bacterial colonies	Bacterial isolate number
2nd nymphal stage	2	LSB2A, LSB2B	2	LSB2C, LSB2D,	1	LSB2E
4th nymphal stage	1	LSB4A	3	LSB4B, LSB4C, LSB4D	2	LSB4E, LSB4F
Adult male	2	LSBM1, LSBM2	2	LSBM3, LSBM4,	2	LSBM5, LSBM6,
Adult female	2	LSBF1, LSBF2,	3	LSBF3, LSBF4, LSBF5	2	LSBF6, LSBF7

Table 2: Morphological and Biochemical characteristics of bacteria associated with the different developmental stages of *Tessaratoma javanica*

Characteristics of isolated bacteria	Bacterial isolates									
	LSB2D, LSBM3, LSBF7, LSB4E	LSBM1, LSBF1, LSB4A	LSBF2, LSB2, LSBM2,	LSBF3, LSB4F,	LSBF5, LSBM4, LSB2A,	LSBF6, LSB4D,	LSB2C, LSB4B,	LSB2E,	LSBM6, LSB4C LSBF4,	LSBM5
Shape	Rod	Rod	Rod	Cocci	Cocci	Rod	Rod	Rod	Rod	Rod
Motility	Non motile	Motile	Motile	Non motile	Non motile	Non motile	motile	motile	Motile	Non motile
Gram reaction	+	+	+	+	+	+	-	-	+	-
Colonies color	Yellow	Creamy White	Off White	Beige	Pale yellow	Pale orange	Dull grey	Light	Dull white	Opaque
Growth in broth medium	sediment	sediment	cloudy	sediment	sediment	cloudy	sediment	sediment	turbidity	turbidity
Biochemical test										
Citrate test	+	+	+	-	-	+	+	-	+	+
Indole test	-	-	-	-	-	-	+	-	-	-
TSI test	+	+	+	+	-	-	-	-	-	-
Catalase test	+	+	+	+	+	+	+	+	+	+
Oxidase test	-	+	-	-	-	-	-	-	-	-
Carbohydrate fermentation										
D-Glucose	+	+	+	+	+	+	+	+	+	+
D-Glucose (gas production)	+	+	_	+	-	-	+	-	-	-
Sucrose	+	+	+	+	+	+	+	+	+	-
Lactose	+	+	-	+	+	-	+	-	+	-
Maltose	+	+	+	+	+	+	+	-	+	-
D-Mannose	+	+	+	+	+	-	+	-	+	-
D-Xylose	+	+	+	-	+	+	+	+	+	-
Identified as	*Microbacterium paraoxydans*	*Bacillus tequilensis*	*Bacillus sp.*	*Staphylococcus arlettae*	*Staphylococcus warneri*	*Microbacterium testaceum*	*Enterobacter*	*Pseudomonas sp.*	*Bacillus subtilis*	*Acinetobacter*

According to the data in Table 1, different developmental stages are indicated by the abbreviation.

LSB code for Litchi stink bug. A, B, C, D, E show the different types of bacterial isolates in the 2nd and 4th nymphal stage. 1,2,3,4,5,6,7 show the different types of bacterial isolates in adult male and female stage. The second nymphal stages are denoted by LSB 2. The Fourth nymphal Stages of the Litchi Stink Bug is represented by LSB 4. Male stage is denoted as LSBM. Female stage denoted as LSBF.

According to the data in Table 2, different developmental stages are indicated by the abbreviation. LSB code for Litchi stink bug. LSBA, B, C, D, E show the different types of bacterial isolates in 2nd and 4th nymphal stages. LSBM & LSBF 1,2,3,4,5,6,7 show the different types of bacterial isolates in adult male and female stages. LSB2 Litchi stink bug second nymphal stage, A, B, C, D, E show the different types of bacterial isolates. The second nymphal stages are denoted by LSB 2. The Fourth nymphal stage of the Litchi Stink Bug is represented by LSB 4. The male stage is denoted as LSBM. Female stage denoted as LSBF. Positive result indicated as Negative result indicated as -

Isolation of bacterial isolates associated with the developmental stages of litchi stink bugs, *Tessaratoma javanica*.

Bacteria linked to various developmental stages, which include the 2nd nymphal stage, 4th nymphal stage, the male adult, and the female adult, were isolated in 3 different cultural media: PYEA (Peptone yeast extract agar, NA (Nutrient agar), and BHI (Brain heart infusion). It was found that bacteria were associated with all developmental stages. The no. of isolates in the 2nd nymphal stage was 5, and no. of isolates in the 4th nymphal stage was 6. The no. of isolates in adult males was 6, and the number of isolates in adult females was 7. The number of isolates was more in the adult female stage, followed by the male and nymphal stages (4th and 2nd stages). The no. of isolates was 7 in PYEA, 10 in NA, and 7 in BHI media. The no. of bacterial isolates was more in Nutrient agar than in PYEA and least found in BHI. A total of 24 bacterial colonies were isolated in 3 different cultural media

Morphological characterization of the bacterial isolates

All the bacterial isolates were Gram-positive except LSB2C, LSB4B, LSB2E, and LSBM5 are Gram-negative. All the bacterial isolates were rod-shaped except LSBF3, LSB4F, LSBF5, LSBM4, and LSB2A; LSBM4 were cocci in shape. Most of the isolates LSB2D, LSBM3, LSBF7, LSB4E, LSBF3, LSB4F, LSBF5, LSBM4, LSB2A, LSBF6, LSB4D, and LSBM5 were non motile but few bacterial isolates like LSBM1, LSBF1, LSB4A, LSBF2, LSB2B, LSBM2, LSB2C, LSB4B, LSB2E, LSBM6, LSB4C, and LSBF4 were motile found in all developmental stages. All the bacterial colonies were of different colors, which include Yellow, Creamy White, Off White, Beige, Pale yellow, Pale orange, Dull grey, Light, Dull white, and Opaque. Most of the bacterial isolates were Sediment types of bacterial growth in broth medium but LSBF2, LSB2B, LSBM2, LSBF6, LSB4D had a cloudy appearance, and LSBM6, LSB4C, LSBF4, and LSBM5 were turbidity in appearance.

Biochemical characterization of bacterial isolates 2nd nymphal stage

The bacterial isolates of 2nd nymphal stage, LSB2B, LSB2C, LSB2D have Positive Citrate test but LSB2A, LSB2E have negative citrate test. All the bacterial isolates have negative indole test except LSB2C that shown positive Indole test. LSB2B and LSB2D have a positive TSI test while rest of all isolates of the 2nd developmental stage have negative TSI tests. All the isolates gave a Positive catalase test, and all the isolates had a negative Oxidase test. D glucose test gave a positive result for all isolates. The D-glucose (gas production) gave a negative reaction for LSB2B, LSB2A and LSB 2E and rest all gave positive reaction. All the isolates gave a positive reaction for Sucrose test. Isolates LSB2B and LSB2E gave a negative result for the Lactose test, the rest all have a positive lactose test. All the isolates gave positive results for the Maltose, D-Mannose, and D-Xylose test except that LSB2E is negative for maltose and D-Mannose.

In the 4th nymphal stage

Isolates LSB4F gave a negative reaction for the Citrate test, but the rest all had positive results. Only Isolates LSB4B have a positive Indole test rest of the isolates gave a negative result. LSB4B, LSB4D, and LSB4C gave a negative TSI result rest all have positive. All the isolates gave positive results for the Catalase test, D-Glucose, and Sucrose test. Only LSB4A has a positive Oxidase test rest all gave negative results. LSB4C and LSB4D gave negative results; all isolates gave positive D-Glucose (gas production) test. All the isolates gave a positive result for

the Lactose test except LSB4D, which had a negative result. All the isolates gave a positive reaction for the D-Mannose test except LSB4D, which had a negative result. In the D-Xylose test, only LSB4F gave a negative result rest all gave a positive reaction.

Adult male stage:

All the isolates gave a positive reaction for the Citrate test, except that LSBM4 gave a negative result. All the isolates gave negative results for the Indole test, and all the isolates gave positive Catalase reaction. LSBM4, LSBM5, and LSBM6 gave a negative TSI reaction, whereas LSBM1, LSBM2, and LSBM3 gave a positive result. All the isolates gave a negative Oxidase test except LSBM1, which had positive results. All isolates gave a positive reaction for the D-Glucose test. LSBMI and LSBM3 have positive results for the D-glucose (Gas production) test rest have negative results. Sucrose, Lactose, Maltose, D-Mannose, and D-Xylose tests gave positive results for all isolates except for LSBM5 gave negative results for all tests, and LSBM2 gave negative results for the Lactose reaction.

Adult female stage:

Isolates LSBF3 and LSBF5 gave negative reaction for the Citrate test, the rest all have positive Citrate test. All the isolates gave a negative reaction for the Indole test and positive for Catalase test. Isolates LSBF4, LSBF5, and LSBF6 gave a negative result for TSI test rest all have positive. Only isolates LSBF1 gave a positive reaction for Oxidase test rest all gave negative result. All the isolates gave positive reaction for the D-Glucose test, Sucrose and Maltose test. Isolate LSBF1, LSBF3 and LSBF7 gave a positive reaction for D-Glucose (gas production) test rest all have negative result. LSBF2 and LSBF6 gave a negative reaction for Lactose test rest all have positive result. Isolate LSBF6 gave a negative reaction for D-Mannose test rest all have positive reaction. LSBF3 gave a negative D-Xylose test, rest all isolate gave positive reaction.

Table 3: Identified cultural bacteria from different developmental stages of *Tessaratoma javanica*

S.N.	Family	Bacteria	2nd nymph	4th nymph	Male	Female
1.	Microbacteriaceae	*Macrobacterium paraoxydans*	+	+	+	+
2.	Bacillaceae	*Bacillus tequilensis*	-	+	+	+
3.	Bacillaceae	*Bacillus sp.*	+	-	+	+
4.	Staphylococcaceae	*Staphylococcus arlettae*	-	+	-	+
5.	Staphylococcaceae	*Staphylococcus warneri*	+	-	+	+
6.	Microbacteriaceae	*Microbacterium testaceum*	-	+	-	+
7.	Enterobacteriaceae	*Enterobacter*	+	+	-	-
8.	Pseudomonadaceae	*Pseudomonas sp.*	+	-	-	-
9.	Bacillaceae	*Bacillus subtilis*	-	+	+	+
10.	Moraxellacea	*Acinetobacter*	-	-	+	-
Total bacterial species			5	6	6	7

Identification of bacterial isolates associated with the developmental stage of the litchi stink bug, *Tessaratoma javanica*

The bacterial community associated with gut of litchi stink bug in different developmental stages isolated by cultural dependent method. All the bacterial isolates were morphologically and biochemically characterized and belongs to various genera different family. A total of 24 different bacterial colonies on the basis of morphology of colony and out of which, 10 different bacterial species belong to 7 different families were identified from different developmental stages (2nd nymphal, 4th nymphal, adult male and adult female) of *Tessaratoma javanica*. The bacteria species viz., *Microbacterium paraoxydans, Bacillus tequilensis, Bacillus* sp.*, Staphylococcus arlettae, Staphylococcus warneri, Microbacterium testaceum, Enterobacter, Pseudomonas* sp.*, Bacillus subtilis and Acinetobacter* were identified to be associated with different developmental stages. Out of total bacterial isolates, five bacterial isolates from 2nd nymphal stage, six from 4th nymphal stages, six from adult male stage and seven from adult female stages were identified. Genera *Microbacterium paraoxydans*, *Bacillus tequilensis,* and *Pseudomonas sp.* were observed in all the developmental stages. It has been also postulated that gut enterobacteria are dispersed into the female reproductive system, where they are subsequently transferred to the eggs, then to fruit during oviposition and finally passed to the fly offspring.[24,25]

DISCUSSION

As far as we are aware, this is the first study of its kind to explore a traditional culturally dependent approach to investigate the cultivable bacterial community in different developmental stage (2nd nymphal stage, 4th nymphal stage, adult male and adult female). The gut which is rich in symbiotic bacterial communities, is the home to the great majority of insect symbiotic bacteria.[26] The gut of various stinkbugs is specifically associated with obligate symbionts that are essential for the plant sap feeding insects.[27] Here we found 24 different bacterial colonies with 10 different bacterial species which include *Microbacterium paraoxydans, Bacillus tequilensis, Bacillus* sp.*, Staphylococcus arlettae, Staphylococcus warneri, Microbacterium testaceum, Enterobacter, Pseudomonas* sp.*, Bacillus subtilis* and *Acinetobacter* were identified to be associated with different developmental stages. Among them *Microbacterium paraoxydans* bacterial species belongs to the family Microbacteriaceae was found in all developmental stages. And the most prominent bacteria were Bacillaceae. The most of the bacterial isolates was found in female stage. The bacillus was one of the most common types of bacterial isolates found in many insects like Armyworm, *Mythimna separata*. These were few bacterial isolates found in the Hemiptera, *Tessaratoma javanica* but a vast range of bacterial diversity exist in *Tessaratoma javanica* is still unknown. *Bacillus* species, part of the Bacillaceae family is frequently isolated from the insect like gut of flies and beetle using culture-dependent methods, indicating its widespread presence in insect microbiome[28] as these bacteria are recognized for their capacity to generate digestive enzymes that facilitate the breakdown of complex carbohydrates, which can be essential for insect that consume plant matter[29]. Therefore, an extensive study to explore the gut bacterial diversity across the developmental stage of litchi stink bug and to understand the biology, host behavior in relation to gut bacterial community are needed using both the cultural dependent and cultural independent with replicates. However, our study significantly supplements to the available information on the cultivable bacterial diversity associated with the different developmental stage of *Tessaratoma javanica.*

CONCLUSION

The potential for scientific and industrial progress through the study of gut bacteria in insects is significant. These microorganisms perform various metabolic functions, such as breaking down complex molecules, producing bioactive compounds, and regulating host physiology. Enhancing our understanding of microbiota-insect interactions can facilitate the isolation and cultivation of these microorganisms, leading to innovations in biotechnology, agriculture, medicine, and environmental sustainability. Future research should focus on improving culture techniques, integrating genomic tools, and exploring symbiotic behaviors to fully harness the capabilities of these microbial communities.

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